

DYNAMICAL SYSTEMS - MA3081

Solutions to Exercise Sheet 11

Exercise 11.1 (Lyapunov exponents of the logistic map).

- a) Consider the logistic map $f(x) = px(1-x)$ on the unit interval $[0, 1]$ with $p \in (1, 3)$. In this case, all initial conditions $x_0 \in (0, 1)$ converge to the fixed point $\frac{p-1}{p}$.

Compute the Lyapunov exponent $l(x_0)$ for $x_0 \in (0, 1)$.

- b) Numerically plot a bifurcation diagram for the logistic map, analogously to Figure 39(a), for $p \in [0, 4]$.
- c) Numerically compute the Lyapunov exponent of the logistic map for all $p \in [0, 4]$. Compare your results with the bifurcation diagram.

Solution.

- a) First note, that in one dimension the definition of the Lyapunov exponent simplifies and we have

$$l(x_0) = \lim_{k \rightarrow \infty} \frac{1}{k} \ln \left(\prod_{i=0}^{k-1} |f'(x_i)| \right) = \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{i=0}^{k-1} \ln (|f'(x_i)|)$$

Now take some initial condition $x_0 \in (0, 1)$. Since $x_k \rightarrow x_* = \frac{p-1}{p}$ and f is C^1 we have $f'(x_k) \rightarrow f'(x_*) = 2 - p$. Assuming that $x_k \neq \frac{1}{2}$ for all k (so $f'(x_k) \neq 0$) and $p \neq 2$, we show that $l(x_0) = \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{i=0}^{k-1} \ln (|f'(x_i)|) = \ln(f'(x_*)) = \ln(|2 - p|)$.

Let $\epsilon > 0$ and take $N \in \mathbb{N}$ such that $|\ln(|f'(x_i)|) - \ln(|2 - p|)| < \frac{\epsilon}{2}$ for all $i \geq N$. Now we have

$$\begin{aligned} & \left| \frac{1}{k} \sum_{i=0}^{k-1} \ln (|f'(x_i)|) - \ln(|2 - p|) \right| \leq \frac{1}{k} \sum_{i=0}^{k-1} \left| \ln (|f'(x_i)|) - \ln(|2 - p|) \right| \\ &= \frac{1}{k} \sum_{i=0}^{N-1} \left| \ln (|f'(x_i)|) - \ln(|2 - p|) \right| + \frac{1}{k} \sum_{i=N}^{k-1} \left| \ln (|f'(x_i)|) - \ln(|2 - p|) \right| \\ &\leq \frac{1}{k} \sum_{i=0}^{N-1} \left| \ln (|f'(x_i)|) - \ln(|2 - p|) \right| + \frac{1}{k} \sum_{i=N}^{k-1} \frac{\epsilon}{2} \\ &\leq \frac{1}{k} \sum_{i=0}^{N-1} \left| \ln (|f'(x_i)|) - \ln(|2 - p|) \right| + \frac{\epsilon}{2}. \end{aligned}$$

Since the first sum is finite, by choosing k large enough we can also achieve that

$$\frac{1}{k} \sum_{i=0}^{N-1} \left| \ln (|f'(x_i)|) - \ln(|2 - p|) \right| \leq \frac{\epsilon}{2}$$

and thus we have $\left| \frac{1}{k} \sum_{i=0}^{k-1} \ln(|f'(x_i)|) - \ln(|2-p|) \right| \leq \epsilon$. Since ϵ was arbitrary, this shows

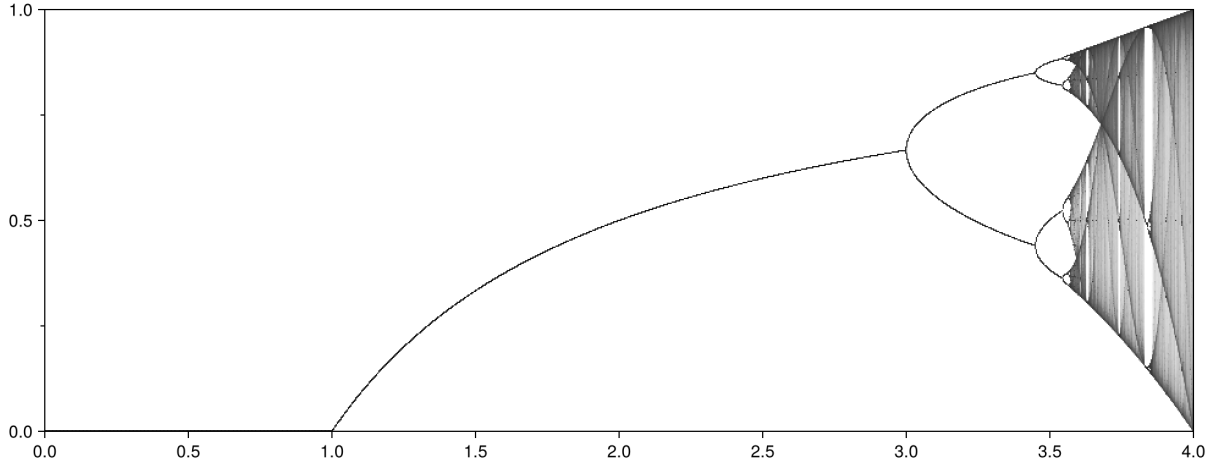
$$l(x_0) = \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{i=0}^{k-1} \ln(|f'(x_i)|) = \ln(|2-p|).$$

Setting $\ln(0) = -\infty$ one obtains in a similar manner that $l(x_0) = -\infty$ if $p = 2$.

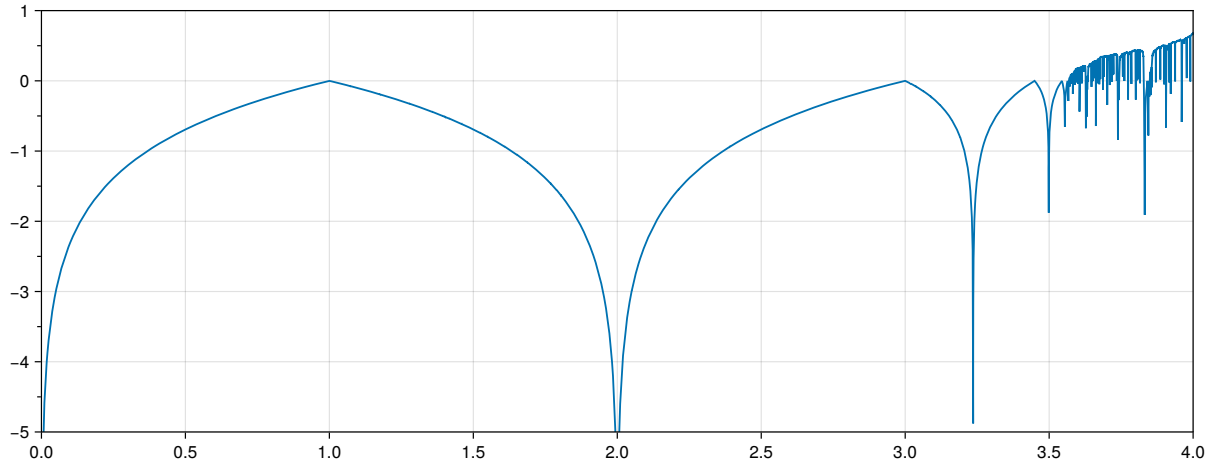
Furthermore, we have $l(x_0) = -\infty$ whenever there is some $n \in \mathbb{N}$ such that $x_n = \frac{1}{2}$, since in this case $\prod_{i=0}^{k-1} |f'(x_i)| = 0$ for all $k \geq n+1$.

Luckily, this is in general not problematic for the numerical computation of the Lyapunov exponent in b) since those critical points form a (Lebesgue-) null set (the point $\frac{1}{2}$ has only countably many preimages under iterates of f). Hence, taking a random point $x_0 \in (0, 1)$ (w.r.t. a uniform distribution), we will “never” hit one of the critical points with $l(x_0) = -\infty$.

b)



c) At around $p \approx 3.57$ the Lyapunov exponent first becomes positive.



□

Exercise 11.2 (Synchronization of Chaos).

Let $f_p(x): [0, 1] \rightarrow [0, 1]$ be defined as

$$f_p(x) = \begin{cases} px & \text{if } x \in [0, 1/2), \\ p(x - 1/2) & \text{if } x \in [1/2, 1], \end{cases}$$

where $p \in (1, 2]$ is a parameter.

- a) Consider the dynamical system that is given by the iteration of f_p . Calculate its Lyapunov exponent.

Disclaimer: We know that f_p is not in C^2 , but you may take Definition 15.10 anyways. Technically, you can define a set $S = \{x \in [0, 1] : f_p^n(x) \notin \{\frac{1}{2}, 1\} \text{ for all } n \in \mathbb{N}_0\}$ and restrict f_p to S .

- b) Now, consider the two-dimensional system

$$\begin{aligned} x_{n+1} &= f_p(x_n) + a(f_p(y_n) - f_p(x_n)), \\ y_{n+1} &= f_p(y_n) + a(f_p(x_n) - f_p(y_n)), \end{aligned} \quad (11.1)$$

with $a \in [0, 1]$. Show that the unit square $[0, 1]^2$ is invariant and calculate both Lyapunov exponents for this system.

- c) Let $(z_n)_{n \in \mathbb{N}}$ be a sequence with $z_n \in [0, 1]$ such that $z_{n+1} = f_p(z_n)$. Show that $(z_n, z_n)^\top$ is a solution of (11.2). Further, determine the transverse stability of this solution by making the ansatz

$$\begin{aligned} x_n &= z_n + \alpha_n, \\ y_n &= z_n - \alpha_n, \end{aligned}$$

to determine an iteration procedure for α_n when $|\alpha_n| \ll 1$. You may assume that locally $f_p(x + h) = f_p(x) + f'_p(x)h$.

Solution.

- a) Note that we have $f'_p(x) = p$ for all $x \in [0, 1]$. Thus, the matrix A_k from Definition 15.10 is given by $A_k = p^k$. Moreover, $A_k^\top A_k = p^{2k} = (p^k)^2$. Thus, $(\lambda^{(k)})^2$ is an eigenvalue of A_k if $\lambda^{(k)} = p^k$. Using formula 15.5 we obtain

$$l_1(x_0) = \lim_{k \rightarrow \infty} \frac{1}{k} \log(\lambda_1^{(k)}) = \lim_{k \rightarrow \infty} \frac{1}{k} \log(p^k) = \lim_{k \rightarrow \infty} \log(p) = \log(p),$$

as the only Lyapunov exponent.

- b) We introduce the notation $\mathcal{F}(x, y)$ for the right-hand side of the two-dimensional system, i.e.,

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} f_p(x_n) + a(f_p(y_n) - f_p(x_n)) \\ f_p(y_n) + a(f_p(x_n) - f_p(y_n)) \end{pmatrix} =: \mathcal{F}(x_n, y_n). \quad (11.2)$$

To show invariance, we assume that $(x, y) \in [0, 1]^2$. Since $f_p: [0, 1] \rightarrow [0, 1]$, we have $f_p(x) \in [0, 1]$ and $f_p(y) \in [0, 1]$. Because $a \in [0, 1]$,

$$\mathcal{F}_1(x, y) = f_p(x) + a(f_p(y) - f_p(x)) = (1 - a)f_p(x) + af_p(y)$$

is a convex combination of two elements in $[0, 1]$. Consequently, $\mathcal{F}_1(x, y) \in [0, 1]$. Similarly, one can show $\mathcal{F}_2(x, y) \in [0, 1]$. This proves invariance.

To obtain the Lyapunov exponents, we calculate

$$D\mathcal{F}(x, y) = p \begin{pmatrix} 1-a & a \\ a & 1-a \end{pmatrix} = SDS^{-1} =: B$$

with

$$D = p \begin{pmatrix} 1 & 0 \\ 0 & 1-2a \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Note that this is independent of x, y . Therefore, the matrix A_k from Definition 15.10 is given by

$$A_k = B^k = SD^kS^{-1} = S \begin{pmatrix} p^k & 0 \\ 0 & (p(1-2a))^k \end{pmatrix} S^{-1}.$$

Due to symmetry of B and thus also of A_k we have

$$A_k^\top A_k = A_k A_k = SD^{2k}S^{-1}.$$

Since the basis change does not affect the eigenvalues of $A_k^\top A_k$, they agree with the eigenvalues of D^{2k} . In particular, they are given by $(\lambda_1^{(k)})^2 = p^{2k}$ and $(\lambda_2^{(k)})^2 = (p(1-2a))^{2k}$. Therefore, we obtain

$$\begin{aligned} l_1(x_0) &= \lim_{k \rightarrow \infty} \frac{1}{k} \log(\lambda_1^{(k)}) = \lim_{k \rightarrow \infty} \frac{1}{k} \log(p^k) = \lim_{k \rightarrow \infty} \log(p) = \log(p), \\ l_2(x_0) &= \lim_{k \rightarrow \infty} \frac{1}{k} \log(\lambda_2^{(k)}) = \lim_{k \rightarrow \infty} \frac{1}{k} \log(|p(1-2a)|^k) = \lim_{k \rightarrow \infty} \log(|p(1-2a)|) = \log(|p(1-2a)|). \end{aligned}$$

c) We have

$$\mathcal{F}(z_n, z_n) = \begin{pmatrix} f_p(z_n) + a(f_p(z_n) - f_p(z_n)) \\ f_p(z_n) + a(f_p(z_n) - f_p(z_n)) \end{pmatrix} = \begin{pmatrix} f_p(z_n) \\ f_p(z_n) \end{pmatrix} = \begin{pmatrix} z_{n+1} \\ z_{n+1} \end{pmatrix}.$$

This confirms that $(z_n, z_n)^\top$ is a solution.

To show the transverse stability of the solution we use the ansatz $x_n = z_n + \alpha_n$ and $y_n = z_n - \alpha_n$. Inserting this ansatz into the iteration formula $(x_{n+1}, y_{n+1})^\top = \mathcal{F}(x_n, y_n)$ yields

$$\begin{aligned} \begin{pmatrix} z_{n+1} + \alpha_{n+1} \\ z_{n+1} - \alpha_{n+1} \end{pmatrix} &= \begin{pmatrix} f_p(z_n + \alpha_n) + a(f_p(z_n - \alpha_n) - f_p(z_n + \alpha_n)) \\ f_p(z_n - \alpha_n) + a(f_p(z_n + \alpha_n) - f_p(z_n - \alpha_n)) \end{pmatrix} \\ &= \begin{pmatrix} f_p(z_n) + f'_p(z_n)\alpha_n + a(f_p(z_n) - f'_p(z_n)\alpha_n - f_p(z_n) - f'_p(z_n)\alpha_n) \\ f_p(z_n) - f'_p(z_n)\alpha_n + a(f_p(z_n) + f'_p(z_n)\alpha_n - f_p(z_n) + f'_p(z_n)\alpha_n) \end{pmatrix} \\ &= \begin{pmatrix} f_p(z_n) + (1-2a)f'_p(z_n)\alpha_n \\ f_p(z_n) - (1-2a)f'_p(z_n)\alpha_n \end{pmatrix}. \end{aligned}$$

Using $f'_p(x) = p$ and $f_p(z_n) = z_{n+1}$ we have

$$\begin{pmatrix} z_{n+1} + \alpha_{n+1} \\ z_{n+1} - \alpha_{n+1} \end{pmatrix} = \begin{pmatrix} z_{n+1} + (1-2a)p\alpha_n \\ z_{n+1} - (1-2a)p\alpha_n \end{pmatrix}.$$

This yields the iteration $\alpha_{n+1} = (1-2a)p\alpha_n$. Thus, the solution $(z_n, z_n)^\top$ is transversally stable if $|(1-2a)p| < 1$ and transversally unstable if $|(1-2a)p| > 1$. $\square$

Exercise 11.3 (Attractors).

Consider the system

$$\begin{aligned}x' &= x - x^3, \\y' &= -y.\end{aligned}$$

- a) Show that the set $Q = [-2, 2] \times [-1, 1] \subset \mathbb{R}^2$ is a trapping region. Determine the corresponding attracting set.
- b) Find an attractor of the system. Is it also a Milnor attractor?

Solution.

- a) One checks that $y' < 0$ on the upper edge of the rectangle, $y' > 0$ on the lower edge, $x' > 0$ on the left edge and $x' < 0$ on the right edge. Hence the flow is everywhere directed inside the set Q , making it a trapping region.

For the corresponding attracting set we obtain $\mathcal{A} = \bigcap_{t \geq 0} \phi_t(Q) = [-1, 1] \times \{0\}$.

- b) The attracting set from a) can be further decomposed in smaller attracting sets, namely the two stable equilibrium points $x_- = (-1, 0)$ and $x_+ = (1, 0)$ (with trapping regions e.g. small balls around each point). Since there cannot be a smaller attracting set than a point, each of the equilibrium points is an attractor.

We find the corresponding basins of attraction as

$$\text{BA}(x_-) = \{(x, y) \in \mathbb{R}^2, x < 0\} \quad \text{and} \quad \text{BA}(x_+) = \{(x, y) \in \mathbb{R}^2, x > 0\}$$

which are clearly sets of positive (Lebesgue-) measure. Again, there can't be a smaller set with a basin of attraction that differs only by a nullset, since we are already reduced to a point. Hence, each stable equilibrium point is also a Milnor attractor.

□